Lesson plan · for the teacher

Databases and SQL for school records: unit check

Data engineering · Level 3 · Unit check

Time	Materials
40 min	Board, chalk, paper. No devices needed.

Learning goals

- From sheets to a database
- Asking questions: SELECT, WHERE, ORDER BY
- Counts and averages: GROUP BY
- Joining tables for reports

Activity: Team quiz

1. Teams (2 min): Make teams of four and give each team a sheet of paper for its answers.
2. Quiz (20 min): Read each question aloud from the quiz sheet; each team agrees on one answer and writes it down.
3. Explain (10 min): Reveal each answer and ask one team to explain why it's right, using an example from the unit.
4. Wrap-up (8 min): Write the question most teams got wrong on the board, and work through it together.

Discussion

- Which school record would you move into a database first?
- Which query would save you the most time?
- Who reads your reports, and what do they need?

Exit ticket

- One query I will write this week, and who will read its answer.

Try it in class tomorrow

Write one real question about your school as a SQL query (on paper is fine), and say who would read its answer and which columns they need.

Worksheet · for pupils

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Name _____ Class _____ Date _____

1 **In a school database, where should each class teacher's phone number be stored?**

◯ Once, in the classes table ◯ In every pupil's row ◯ In every mark row

2 **Match each part of SQL to its job.**

SELECT	• • • •	Which groups come back
WHERE	• • • •	One row per group
GROUP BY	• • • •	Which rows come back
HAVING	• • • •	Which columns come back

3 **Put the parts of a query in the order SQL needs them.**

☐ LIMIT
☐ ORDER BY
☐ GROUP BY
☐ WHERE
☐ FROM
☐ SELECT

4 **This should list 6B's pupils, but returns nothing. Which line is wrong?**

Circle the line with the mistake. Write it correctly:

```
1  SELECT name
2  FROM pupils
3  WHERE class = '6b';
```

5 **A class has 30 pupils; 2 missed the test, so their marks are NULL. What do COUNT(*) and COUNT(mark) return for the class?**

◯ 30 and 28 ◯ 28 and 28 ◯ 30 and 30